

Chapter 6 Processing of Ceramic Materials

5.1 Raw Materials for Traditional Ceramics

5.2 Processing of Traditional Ceramics

5.3 Raw Materials for Advanced Ceramics

5.4 Characterization of Ceramic Powders

5.5 Processing of Advanced Ceramics

Ceramics Processing Overview

- ✦ Traditional ceramics are made from minerals occurring in nature
 - 📄 Products: pottery, porcelain, bricks, and cement
- ✦ Advanced ceramics are made from synthetically produced raw materials, i.e. Alumina (Al_2O_3), Carbon Boron Nitride (CBN), Titanium Carbide (TiC), Tungsten Carbide (WC)
 - 📄 Products: cutting tools, artificial bones, nuclear fuels, substrates for electronic circuits
- ✦ Starting material for these products is powder

Ceramics Processing Overview

✦ For traditional ceramics

- ☞ Powders are mixed with water to bind them together and achieve proper consistency for shaping

✦ For advanced ceramics

- ☞ Substances other than water are used as binders during shaping

✦ After shaping, green part is *fired* (sintered)

- ☞ Function is the same as in PM - to effect a solid state reaction that bonds the particles into a hard mass

5.1 Raw Materials for Traditional Ceramics

Preparation of Raw Materials in Traditional Ceramics Processing

- ✦ Most shaping processes for traditional ceramics require the starting material to be a plastic paste
 - 📄 This paste is comprised of fine ceramic powders mixed with water
- ✦ The starting raw ceramic material usually occurs in nature as rocky lumps
 - 📄 Purpose of the preparation step is to reduce the rocky lumps to powder

Raw Materials for Traditional Ceramics

Naturally occurring minerals

1. Silica (SiO_2): Major source- Sand

2. Silicates:

a. Feldspar (composition from $\text{NaAlSi}_3\text{O}_8$ and KAlSi_3O_8 to $\text{CaAl}_2\text{Si}_2\text{O}_8$)

b. Clay & Kaolin ($\text{Al}_2[\text{Si}_2\text{O}_5][\text{OH}]_4$),

c. Mica: consists of 37 minerals, phyllosilicates;

d. Mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$)

Feldspar 70% used for glass

Kaolin Used in fine china, paper, and rubber

Mica >200,000 t of low-quality mica used each year

Mullite 600,000 t are used each year for refractory furnace blocks

Products:

Whiteware, structural clay products,
brick and tile, refractory, abrasive,
and cement.

<i>Mineral</i>	<i>Chemical formula</i>
Muscovite	$\text{KAl}_2(\text{Si}_3\text{Al})\text{O}_{10}(\text{OH})_2$
Phlogopite	$\text{KMg}_3(\text{AlSi}_3\text{O}_{10})(\text{OH},\text{F})_2$
Paragonite	$\text{NaAl}_2(\text{Si}_3\text{Al})\text{O}_{10}(\text{OH})_2$
Biotite	$\text{K}(\text{Mg},\text{Fe})_3(\text{Al},\text{Fe})\text{Si}_3\text{O}_{10}(\text{OH},\text{F})_2$
Lepidolite	$\text{K}(\text{Li},\text{Al})_3(\text{Al},\text{Si})_4\text{O}_{10}(\text{F},\text{OH})_2$
Zinnwaldite	$\text{KLiFeAl}(\text{AlSi}_3)\text{O}_{10}(\text{F},\text{OH})_2$
Margarite	$\text{CaAl}_2(\text{Al}_2\text{Si}_2\text{O}_{10})(\text{OH})_2$
Clintonite	$\text{Ca}(\text{Mg},\text{Al})_3(\text{Al}_3\text{Si})\text{O}_{10}(\text{OH})_2$

Comminution

Reducing particle size in ceramics processing by using mechanical energy in various forms such as impact, compression, and attrition

- ✦ Comminution techniques are most effective on brittle materials such as cement and metallic ores
- ✦ Two general types of comminution operations:
 1. Crushing
 2. Grinding

Crushing

Reduction of large lumps from the mine to smaller sizes for subsequent further reduction

- ✦ Several stages may be required (e.g., primary crushing, secondary crushing)

- 📄 Reduction ratio in each stage in the range 3 to 6

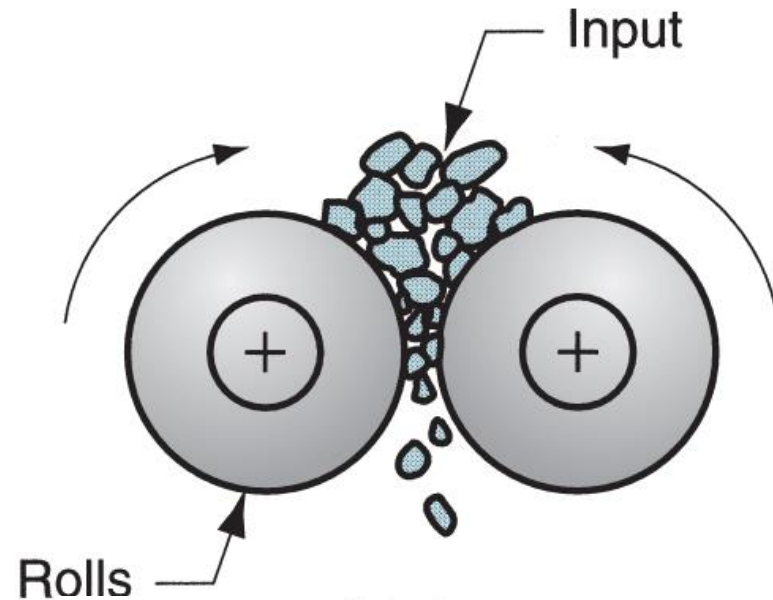
- ✦ Crushing of minerals is accomplished by

- 📄 Compression against rigid surfaces or Impact against surfaces

Crusher

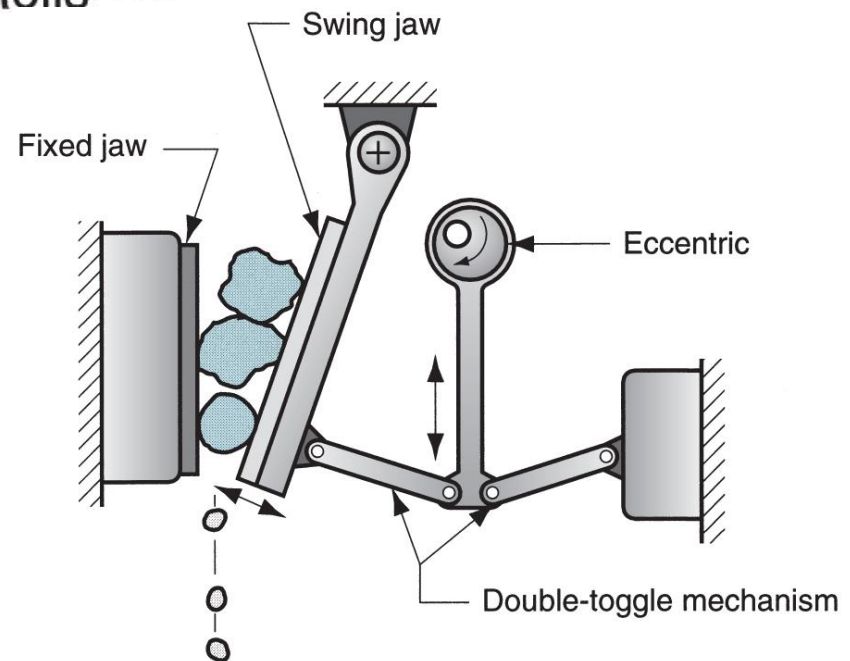
Roll Crusher

- ✦ Ceramic lumps are squeezed between rotating rolls



Jaw Crusher

- ✦ Large jaw toggles back and forth to crush lumps against a hard, rigid surface



Grinding

In the context of comminution, grinding refers to the reduction of small pieces after crushing to fine powder

- ✪ Accomplished by abrasion, impact, and/or compaction by hard media such as balls or rolls

- ✪ Examples of grinding include:

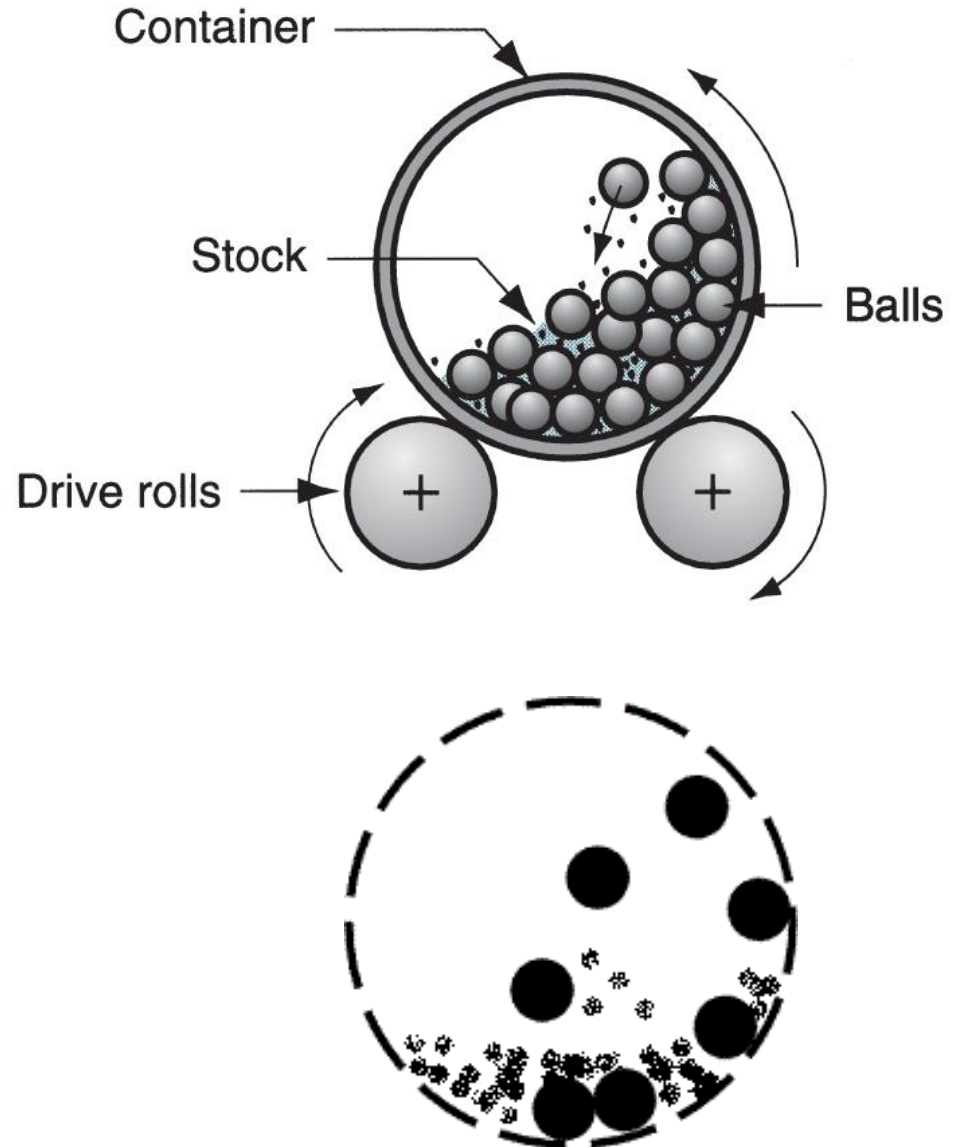
 - 📄 Ball mill

 - 📄 Roller mill

 - 📄 Impact grinding

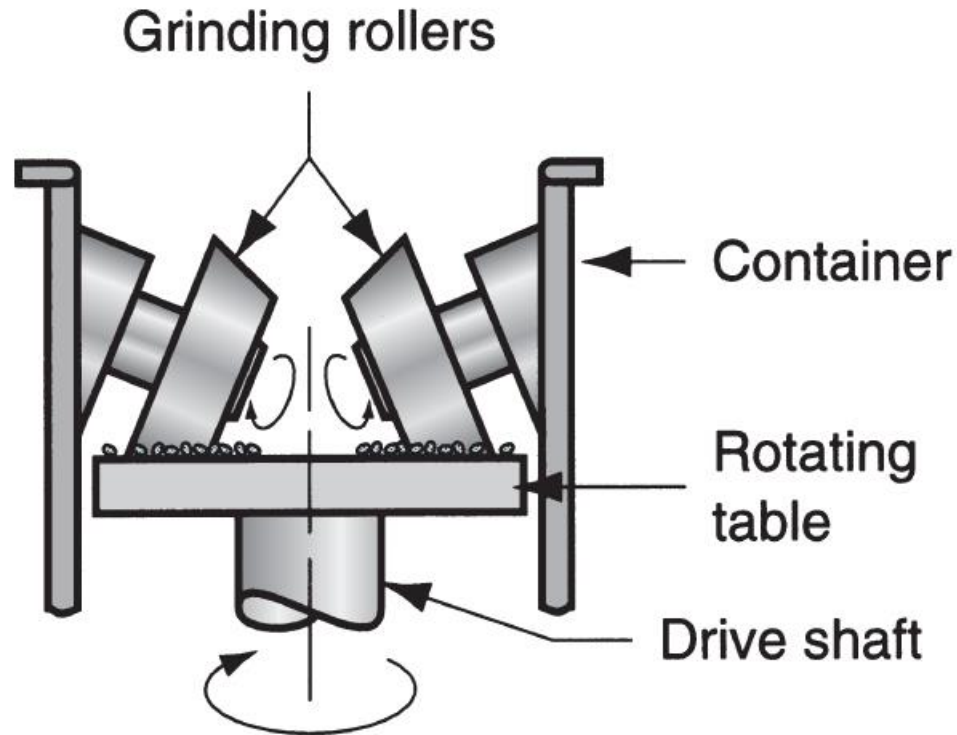
Ball Mill

- ✦ Hard spheres mixed with stock are rotated inside large cylindrical container
- ✦ Mixture is carried upwards in container as it rotates, then dropped by gravity to accomplish grinding action



Roller Mill

- ✦ Stock is compressed against flat horizontal table by rollers riding on the table surface



Main Ingredients of Ceramic Paste

1. Clay

- 📄 Chemistry = hydrous aluminum silicates
- 📄 Usually the main ingredient because of ideal forming characteristics when mixed with water

2. Water

- 📄 Creates clay-water mixture with good plasticity for shaping

Additional Ingredients of Ceramic Paste

3. Non-plastic raw materials

- ☞ Such as alumina and silica
- ☞ Purpose is to reduce shrinkage in drying and firing but also reduces plasticity during forming

4. Other ingredients

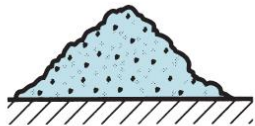
- ☞ Such as fluxes that melt (vitrify) during firing and promote sintering
- ☞ Wetting agents to improve mixing of ingredients

5.2 Processing of Traditional Ceramics

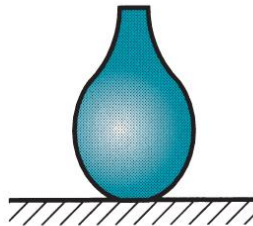
Processing Overview for Traditional Ceramics

- ✦ Condition of powders and part during
- (1) Preparation of raw materials,
 - (2) Shaping,
 - (3) Drying,
 - (4) Firing

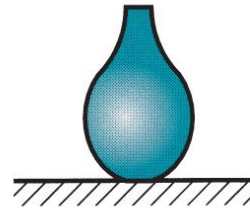
The more water in the mixture, the easier to form but cracking during sintering and drying



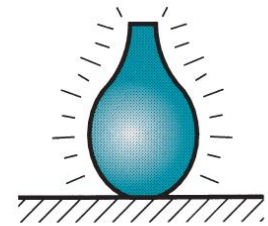
(1) Preparation of powders



(2) Shaping of wet clay

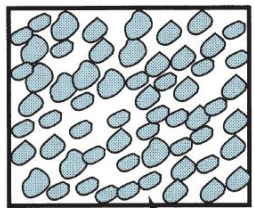


(3) Drying



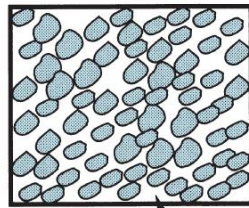
(4) Firing

Loose powders



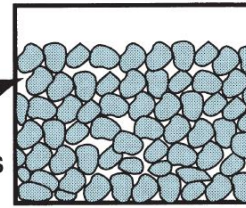
Air

Clay and water



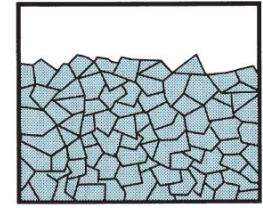
Water

Dried clay



Air pores

Fired clay



Shaping Processes

❖ Slip casting (粉浆浇铸)

📄 The clay-water mixture is a slurry

❖ Plastic forming methods

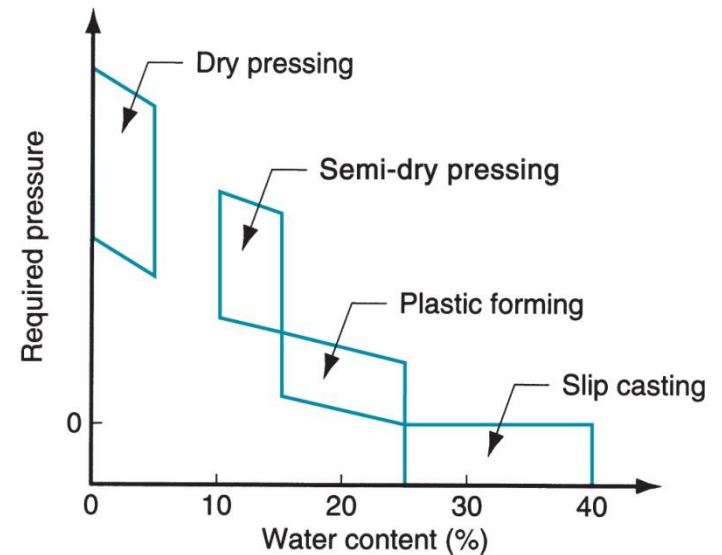
📄 The clay is plastic

❖ Semi-dry pressing

📄 The clay is moist but has low plasticity

❖ Dry pressing

📄 The clay is basically dry (less than 5% water) and has no plasticity



Slip Casting

Suspension of ceramic powders in water, called a *slip*, is poured into porous plaster of paris mold

- ✦ Water from the mix is absorbed into the plaster to form a firm layer of clay at the mold surface

- ✦ Slip composition is 25% to 40% water

- ✦ Two principal variations:

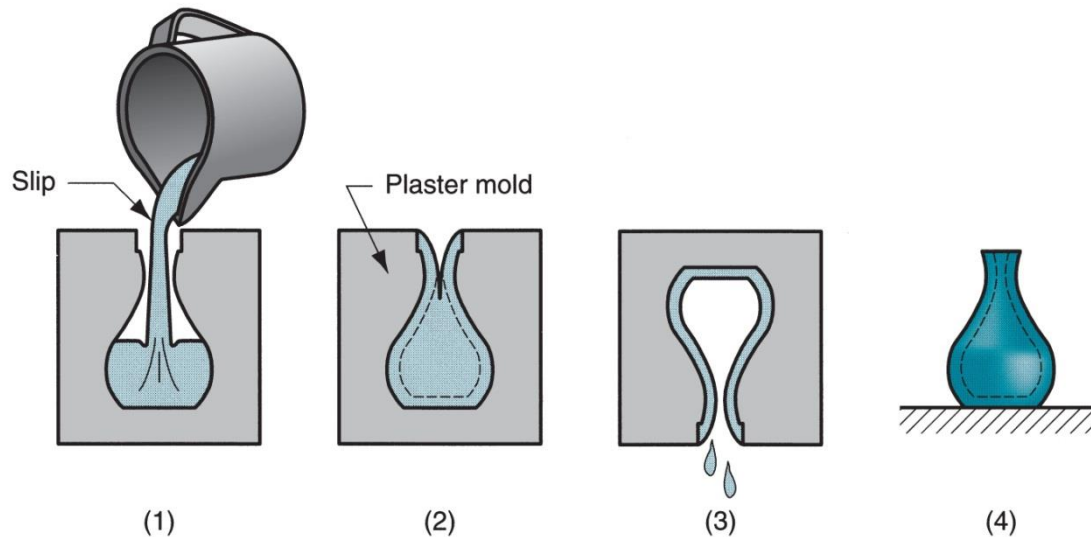
- 📄 Drain casting - mold is inverted to drain excess slip after semi-solid layer has formed

- 📄 Solid casting - adequate time is allowed for entire body to become firm



Drain Casting

- ✎ (1) Slip is poured into mold cavity, (2) water is absorbed into plaster mold to form a firm layer, (3) excess slip is poured out, and (4) part is removed from mold



Overview of Plastic Forming

- ✚ Starting mixture must have a plastic consistency
 - 📄 Composition 15% to 25% water
- ✚ Variety of manual and mechanized methods
 - 📄 Manual methods use clay with more water because it is more easily formed
 - 📄 Mechanized methods generally use clay with less water so starting clay is stiffer
- ✚ Hand modeling (manual method)
- ✚ Jiggering (mechanized method)
- ✚ Plastic pressing (mechanized method)
- ✚ Extrusion (mechanized method)

Hand Modeling

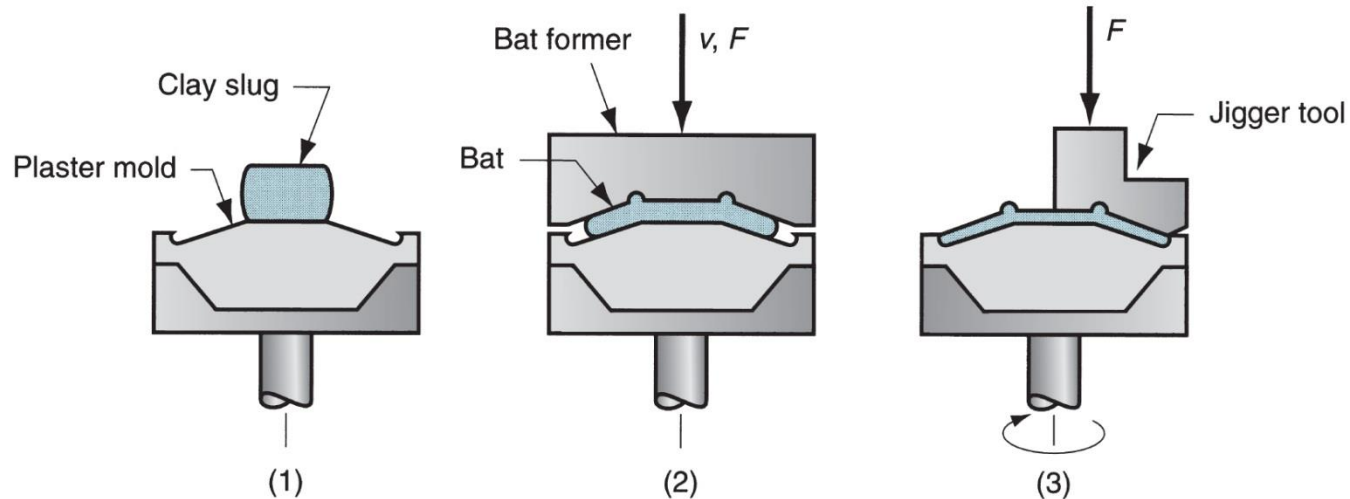
Fabrication of ceramic product by manipulating plastic clay into desired geometry

- ✎ Hand molding - a mold or form is used to define portions of the part geometry
- ✎ Hand throwing on a potter's wheel (陶工旋盘)
 - 📄 Potter's wheel - a round table that rotates on a vertical spindle
 - 📄 Products of circular cross section can be formed by throwing and shaping the clay, sometimes using a mold to provide the internal shape



Jiggering

- 📌 (1) Wet clay slug is placed on a convex mold; (2) batting; and (3) a jigger tool imparts the final product shape



Plastic Pressing

Forming process in which plastic clay slug is pressed between upper and lower molds

- ✦ Molds are made of porous material, so when a vacuum is drawn on the backs of the mold halves, moisture is removed from the clay
- ✦ Mold sections are then opened, using positive air pressure to prevent sticking of part in the mold
- ✦ Advantages: higher production rate than jiggering and not limited to radially symmetric parts

Extrusion

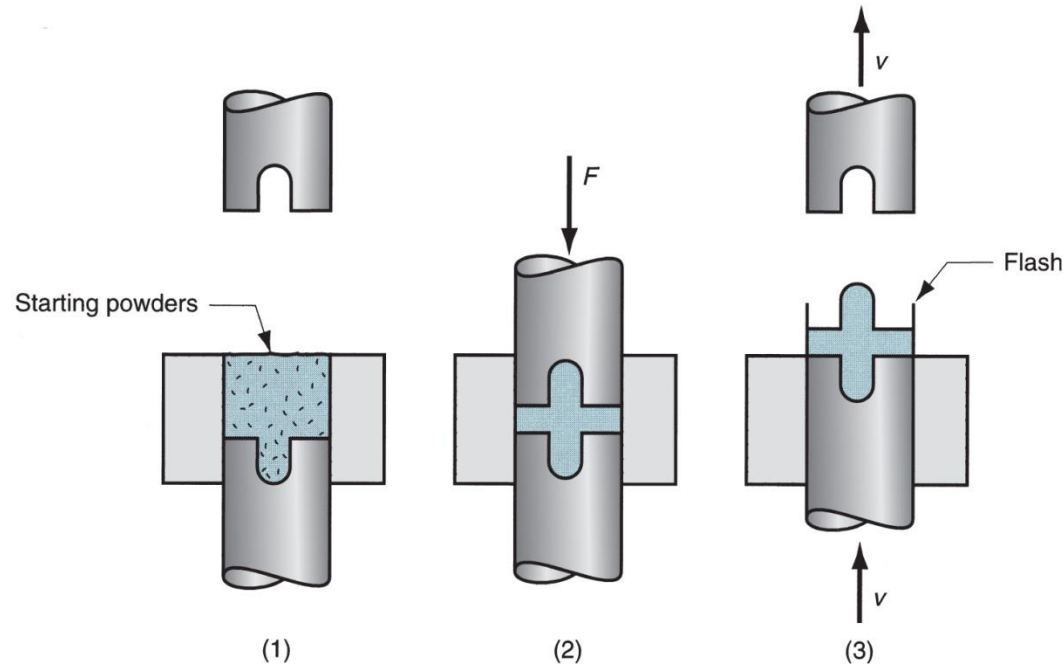
Compression of clay through die orifice to produce long sections of uniform cross section, which are then cut to required piece length

- ✎ Equipment utilizes a screw-type action to assist in mixing the clay and pushing it through die opening
- ✎ Products: hollow bricks, shaped tiles, drain pipes, tubes, and insulators
- ✎ Also used to make starting slugs for jiggering and plastic pressing



Semi-dry Pressing

- ✎ (1) Moist powder deposited in die cavity, (2) press using high pressure, and (3) die sections opened and part ejection



Dry Pressing

- ✚ Process sequence similar to semi-dry pressing
 - 📄 Except water content of starting mix is $< 5\%$
- ✚ Dies made of hardened tool steel or cemented carbide to reduce wear due to abrasive dry clay
- ✚ No drying shrinkage occurs
 - 📄 Drying time is eliminated and good accuracy is achieved in final product
- ✚ Products: bathroom tile, electrical insulators, refractory brick, and other simple geometries

Clay Volume vs. Water Content

- ✪ Water plays an important role in most of the traditional ceramics shaping processes
 - 📄 Thereafter, it has no purpose and must be removed from the clay piece before firing
- ✪ Shrinkage is a problem during drying because water contributes volume to the piece, and the volume is reduced when it is removed

Drying

Drying process occurs in two stages

- ✦ Stage 1 - drying rate is rapid as water evaporates from surface into surrounding air and water from the interior migrates by capillary action to the surface to replace it
 - 📄 This is when volumetric shrinkage occurs, with the risk of warping and cracking

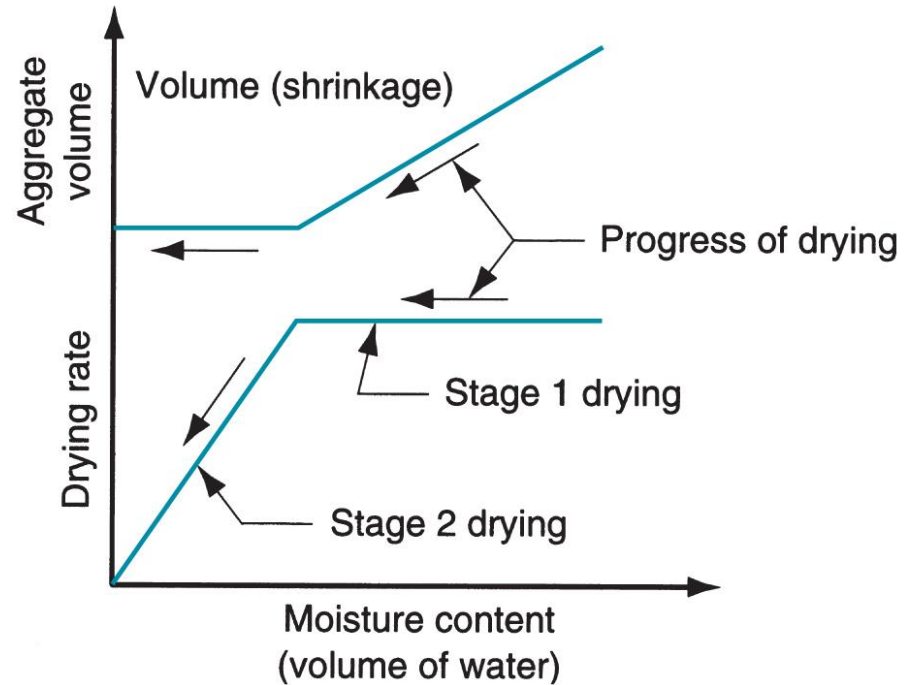
- ✦ Stage 2 - moisture content has been reduced to where the ceramic grains are in contact
 - 📄 Little or no further volumetric shrinkage

Drying Rate and Volume Reduction

✦ Typical drying rate curve and associated volume reduction for a ceramic body

✦ Drying rate in second stage is depicted as a straight line

📄 It is sometimes concave or convex



Firing of Traditional Ceramics

Heat treatment process to sinter the ceramic material

- ✚ Performed in a furnace called a *kiln* (窑炉)
- ✚ Bonds are developed between ceramic grains
 - 📄 This is accompanied by densification and reduction of porosity
 - 📖 Therefore, additional shrinkage occurs in the polycrystalline material in addition to that which has already occurred in drying
- ✚ In firing of traditional ceramics, a glassy phase forms among the crystals that acts as a binder

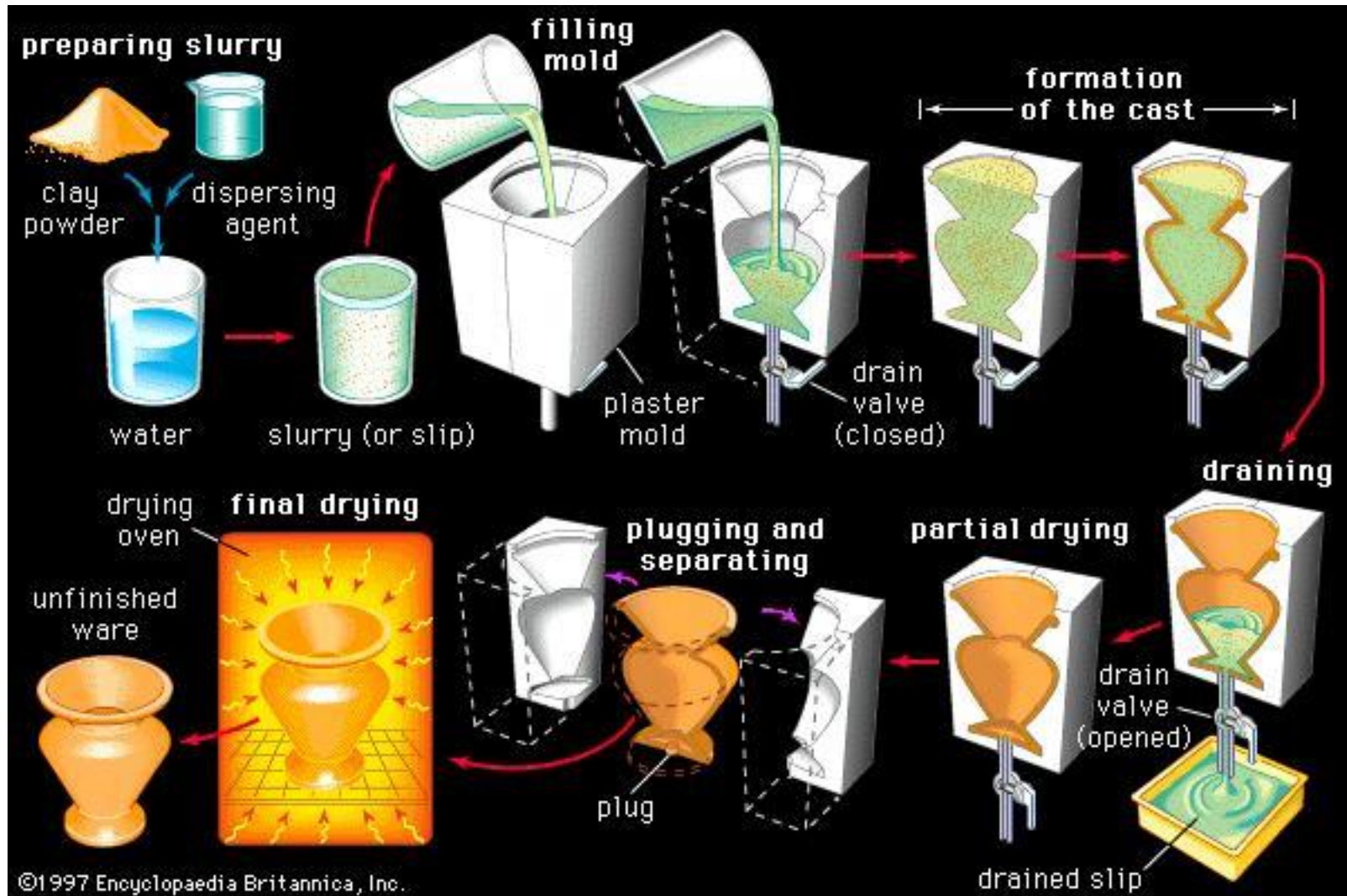
Glazing

Application of a ceramic surface coating to make the piece more impervious to water and enhance its appearance

- 📌 Usual processing sequence with glazed ware:
 1. Fire the piece once before glazing to harden the body of the piece
 2. Apply glaze
 3. Fire the piece a second time to harden glaze

Traditional Ceramic Processing_Summary

Beneficiation → Blending → Forming (Plastic forming/Slip casting) → Firing → Vitrification → Finishing



5.3 Raw Materials for Advanced Ceramics

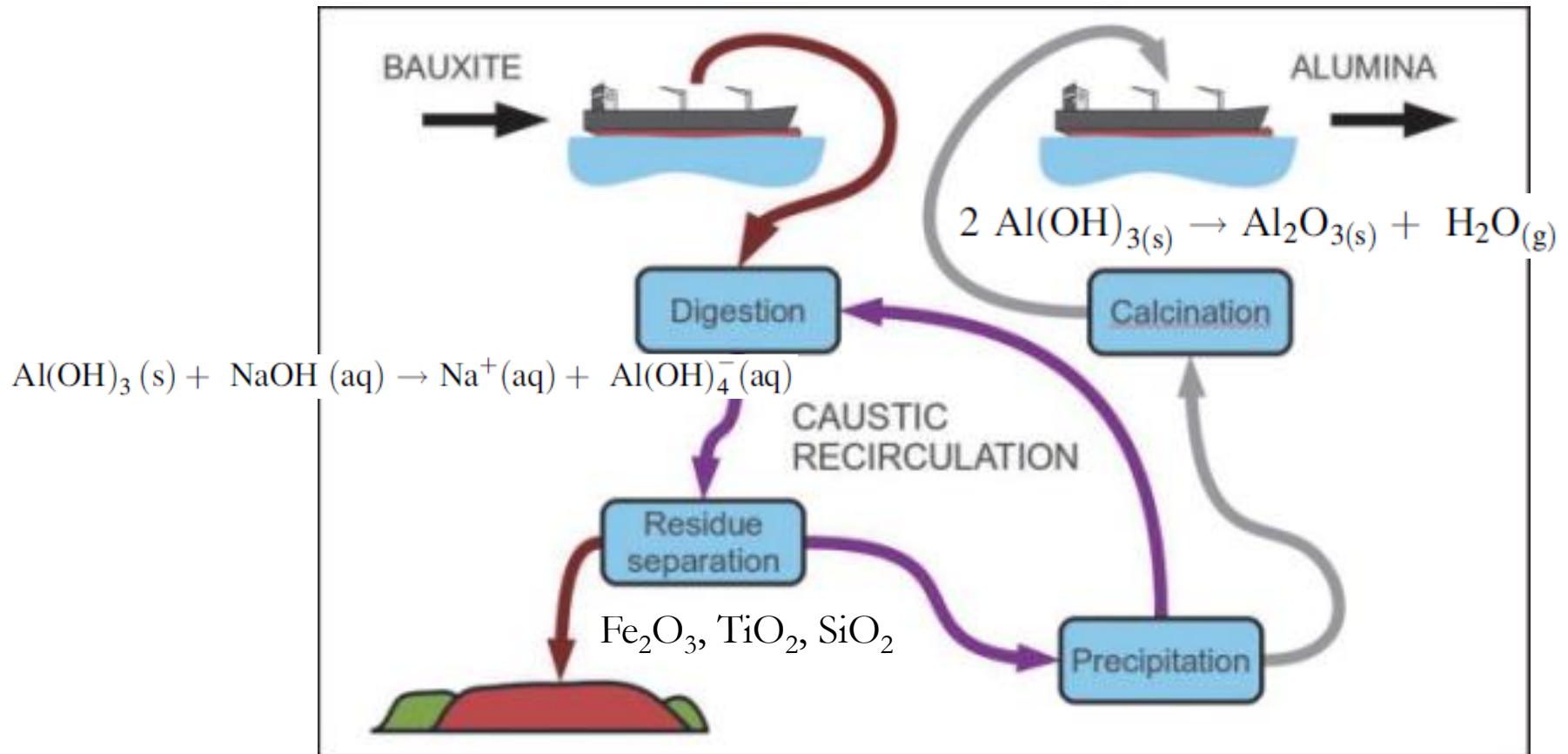
Preparation of Starting Materials

- ✦ Strength requirements are usually much greater for advanced ceramics than for traditional ceramics
- ✦ Starting powders must be smaller and more uniform in size and composition, since the strength of the resulting ceramic product is inversely related to grain size
 - 📄 Greater control over the starting powders is required
 - 📄 Powder preparation includes mechanical and chemical methods

The Raw Material for Alumina

Aluminum oxide (Al_2O_3 , alumina, corundum) : the most widely used inorganic chemical for ceramics and is produced from the mineral bauxite using the Bayer process.

Bauxite: a mixture of hydrated aluminum oxide with iron oxide (Fe_2O_3), silica (SiO_2), and titania (TiO_2) impurities.



The Raw Material for MgO/ZrO₂

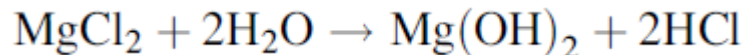
MgO

1) Magnesite → Crushing → Screening → Magnetic separation → froth floatation → calcified (800-900° C) → sintered/dead burned → calcified (1700° C) → MgO

2) Seawater: $\text{Mg}^{2+}(\text{aq}) + 2(\text{OH})^{-}(\text{aq}) \rightarrow \text{Mg}(\text{OH})_2(\text{s})$

Washing → filtering → calcified → MgO

3) Brines: Decomposition of MgCl₂ at 600-800° C



ZrO₂

Raw Material: Zircon, ZrSiO₄;

1) At T > 1750° C: ZrSiO₄ → SiO₂ + ZrO₂;

2) Injection of zircon sand into a plasma (at T > 6,000° C) → dissociation and melting. The zirconia solidifies first, in the form of dendrites, and the silica solidifies as a glassy coating on the zirconia. Remove silica by leaching in boiling sodium hydroxide solution. The residue is washed and the zirconia removed by centrifuging.

3) Electric arc melting of zircon at 2,100–2,300° C.

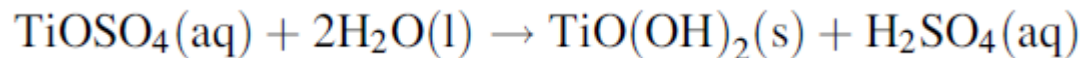
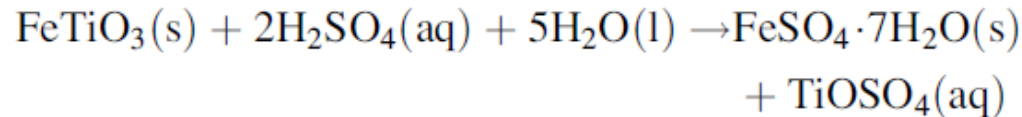
The Raw Material for ZnO and TiO₂

ZnO

- 1) Oxidation of Metal Zn in air;
- 2) Reduction of Sphalerite (ZnS) with C and CO

TiO₂

- 1) A constituent of igneous rocks;
- 2)



Calcine at 1000°C to produce Titania (TiO₂);

The Raw Material for Advanced Nonoxide Ceramics

- 1) Metal + Non-metal at high T ;
- 2) Reduce the oxide with Carbon at high T and then react it with nonmetal.

SiC	Abrasives, harsh-environment electronic packing
TiC	Bearings, cutting tools
AlN	Electronic packaging, crucibles
Si ₃ N ₄	Future gas-turbine and diesel engine components
ZrB ₂	Crucibles and thermowell tubes (steel)
WC	Abrasives, cutting tools
C	Graphite: solid lubricant; diamond: abrasive

SiC



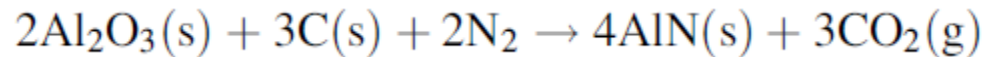
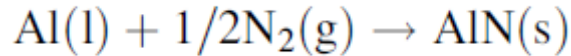
Pure silica sand and carbon in the form of finely ground coke is built up around a carbon conductor within a brick electrical resistance-type furnace. A furnace run can last several days, during which temperatures vary from 2200°C to 2700°C.

The Raw Material for Advanced Nonoxide Ceramics

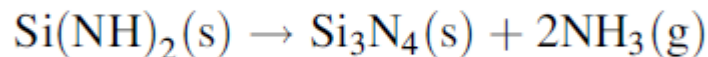
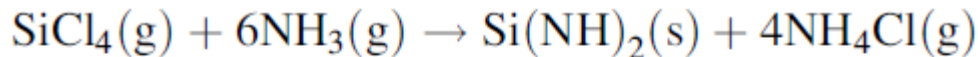
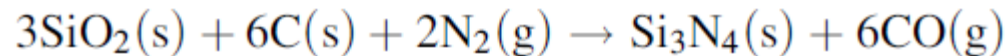
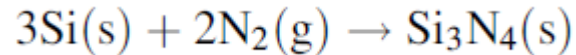
TiC

- 1) Carbothermal reduction of TiO_2 ;
- 2) Direct reaction between element Ti and O_2 .

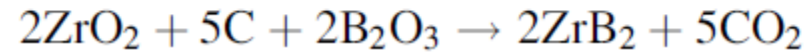
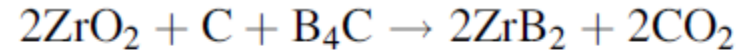
AlN



Si_3N_4



ZrB_2



Making Powders

- Mechanical;
- Chemical;
- Vapor-Phase;

Desirable powder characteristics for advanced ceramics

<i>Powder characteristic</i>	<i>Desired property</i>
Particle size	Fine ($<1\ \mu\text{m}$)
Particle size distribution	Narrow
Particle shape	Spherical or equiaxed
State of agglomeration	No agglomeration or soft agglomerates
Chemical composition	High purity
Phase composition	Single phase

- **Primary particles**—The smallest clearly identifiable unit in the powder. Primary particles may be crystalline or amorphous and cannot easily be broken down into smaller units.
- **Agglomerates**—Clusters of bonded primary particles. Soft agglomerates are easily broken up; hard agglomerates, because of the stronger interparticle bonds, are more difficult. Hard agglomerates should be avoided in ceramic powder processing as much as possible.
- **Particles**—General term applied to both primary particles and agglomerates. Some of the techniques that we refer to in the next section often measure particle size, with no distinction between agglomerates and primary particles.
- **Granules**—Large agglomerates, usually 0.1–1.0 mm in diameter, that are formed by the addition of a granulating agent (e.g., a polymer binder). The mixture is tumbled, producing large, nearly spherical granules that flow freely and can be used for filling complex molds and in automated processes.
- **Flocs**—Clusters of particles in a liquid suspension held together electrostatically.
- **Colloids**—Very fine particles (can be as small as 1 nm in diameter); held in fluid suspension by Brownian motion. Consequently, colloidal particles settle very slowly.
- **Aggregates**—Coarse constituents, $>1\ \text{mm}$, in a mixture. The important example is the addition of gravel to cement to make concrete. In early concrete structures, such as the Pantheon in Rome, pumice was used as aggregate.

Powders by Precipitation Technique

Weigh reactants



Prepare reactant solution



Control pH of solution



Centrifuge the solution



Wash the precipitate



Dry the precipitate



Synthesis of HA by Precipitation Method

Example: Hydroxyapatite (HA) $10\text{Ca}^{2+} + 6\text{PO}_4^{3-} + 2\text{OH}^- \leftrightarrow \text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$

Solution 1#:

500 ml 0.2 M $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$

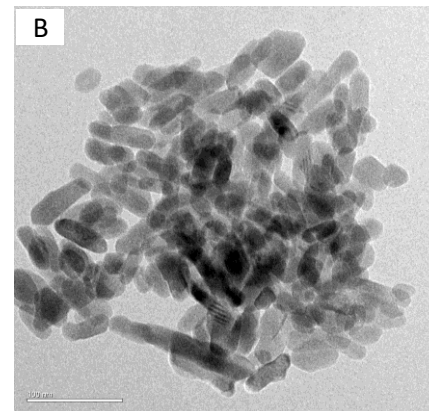
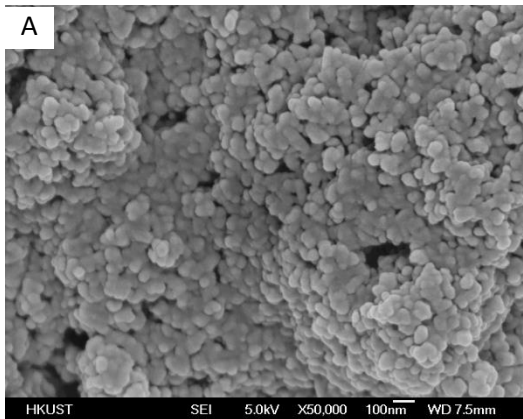
Solution 2#:

500 ml 0.12 M Na_2HPO_4

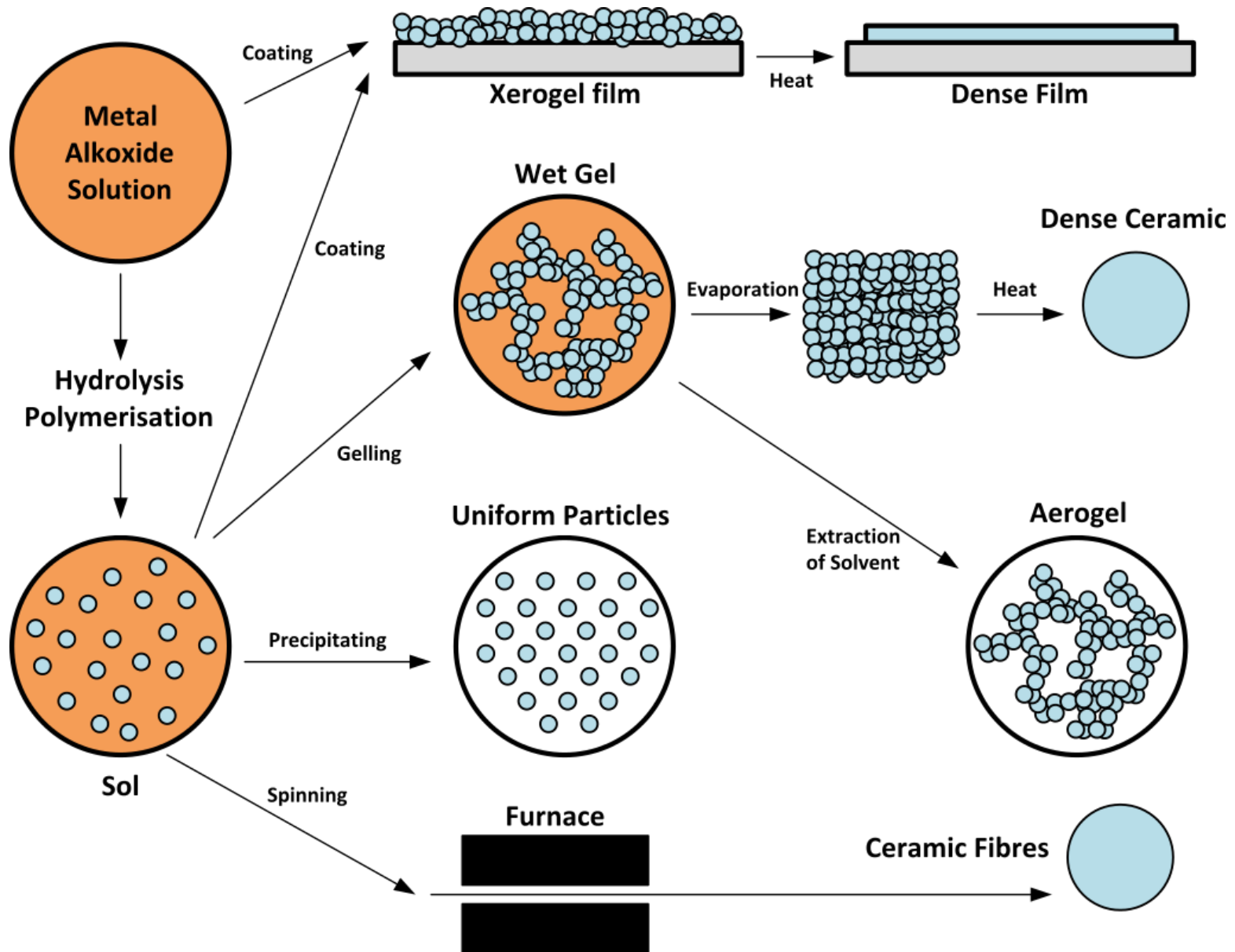
1# dropwise added into 2#, control pH by NaOH solution

Magnetically stirred for $95^\circ\text{C} + 5\text{d}$

Centrifugated, washed with water and dried at 100°C for 10h.

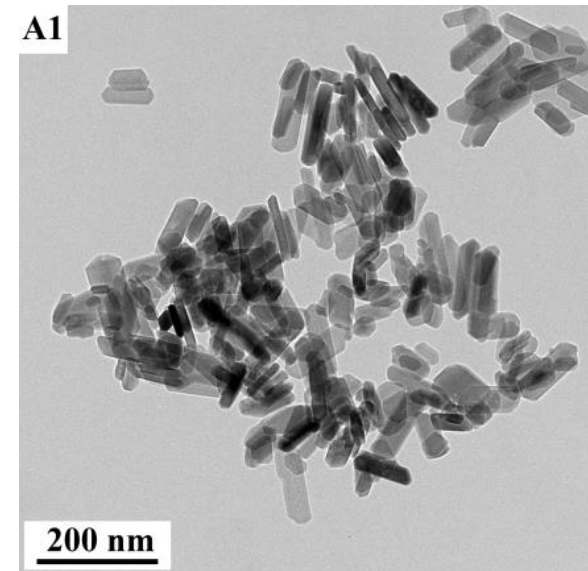
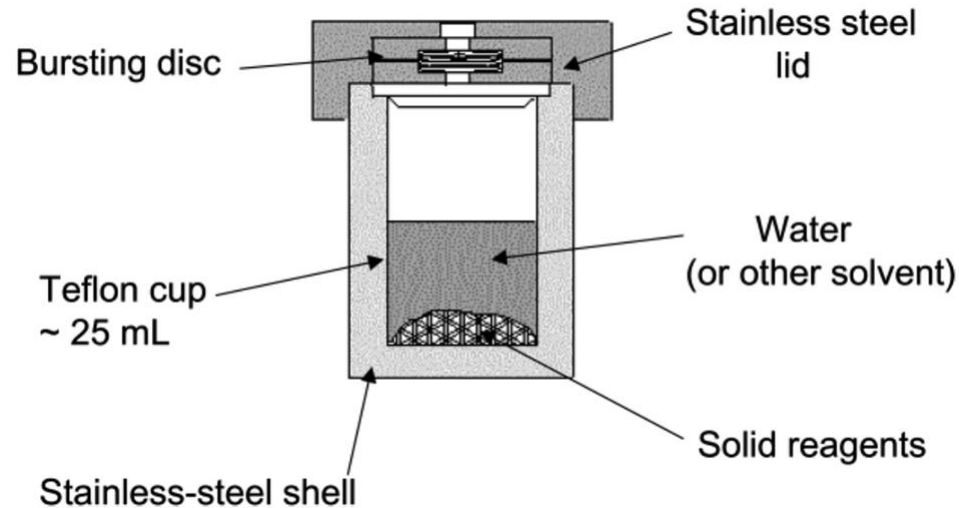


Powders by Sol-Gel Processing



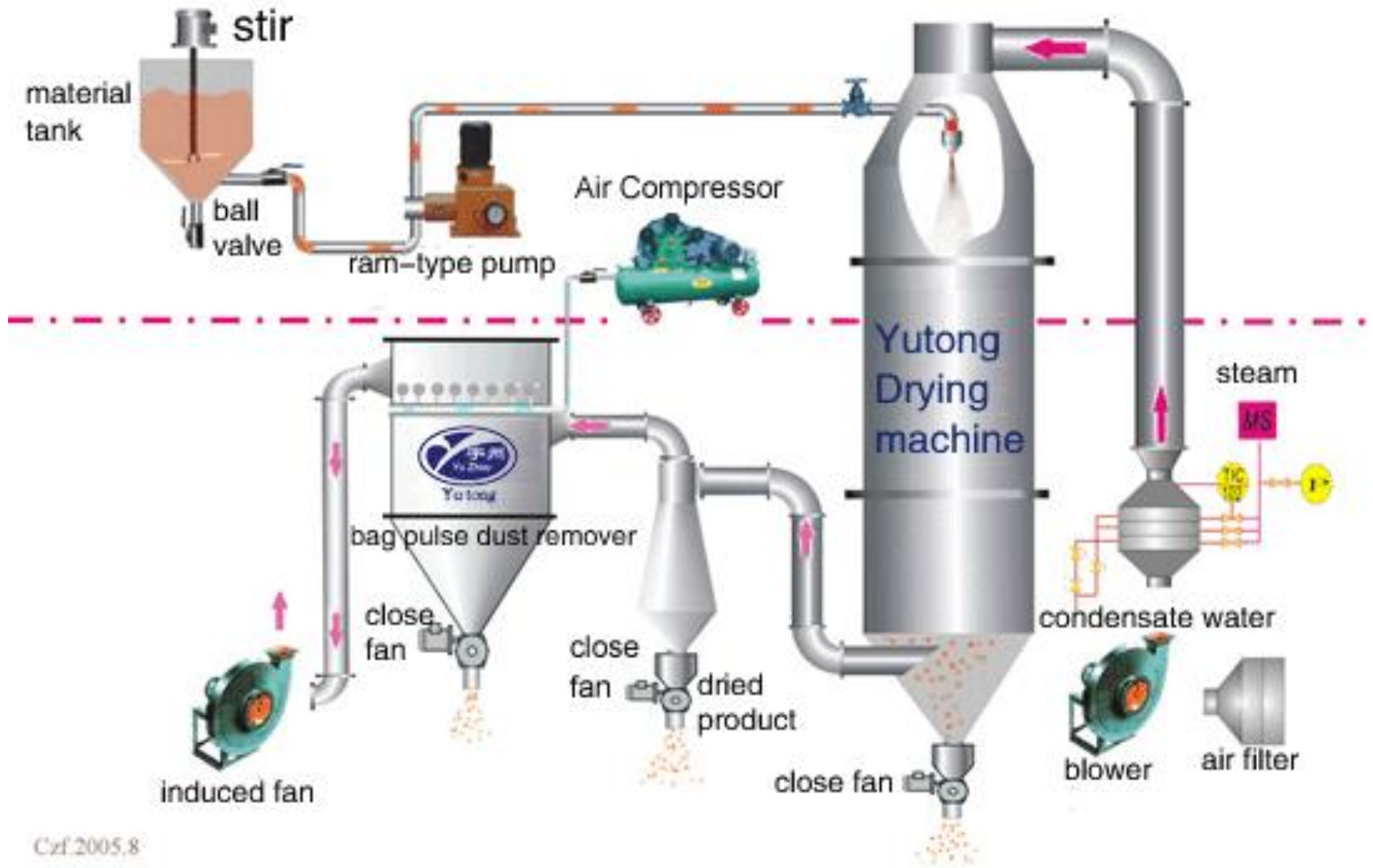
Powders by Hydrothermal synthesis

Hydrothermal synthesis includes the various techniques of crystallizing substances from high-temperature aqueous solutions at high vapor pressures.



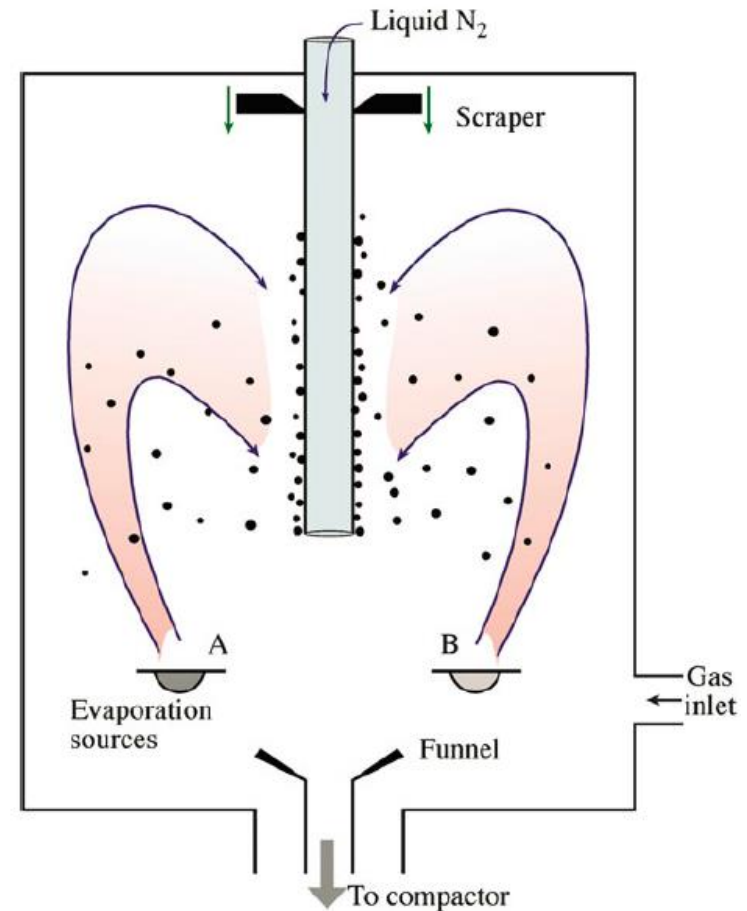
Powder by Spray drying

Spray drying is a method of producing a dry powder from a liquid or slurry by rapidly drying with a hot gas.



Powder by Vapor-Phase Reactions

- High purity;
- Discrete and nanoaggregated particles;
- Nanoparticles with narrow size distribution;
- Versatility in producing powders of oxides and nonoxides.



5.4 Characterization of Ceramic Powder

Powder Composition

- Information related to phases present in the powder or polycrystalline sample $\Rightarrow$ phase analysis
- Phase identification commonly conducted via x-ray diffraction
- Advantages of x-ray diffraction

Rapid

Non-destructive

Does not require extensive sample preparation

X-ray Diffraction

- **Qualitative Analysis**

Phase Identification

- **Quantitative Analysis**

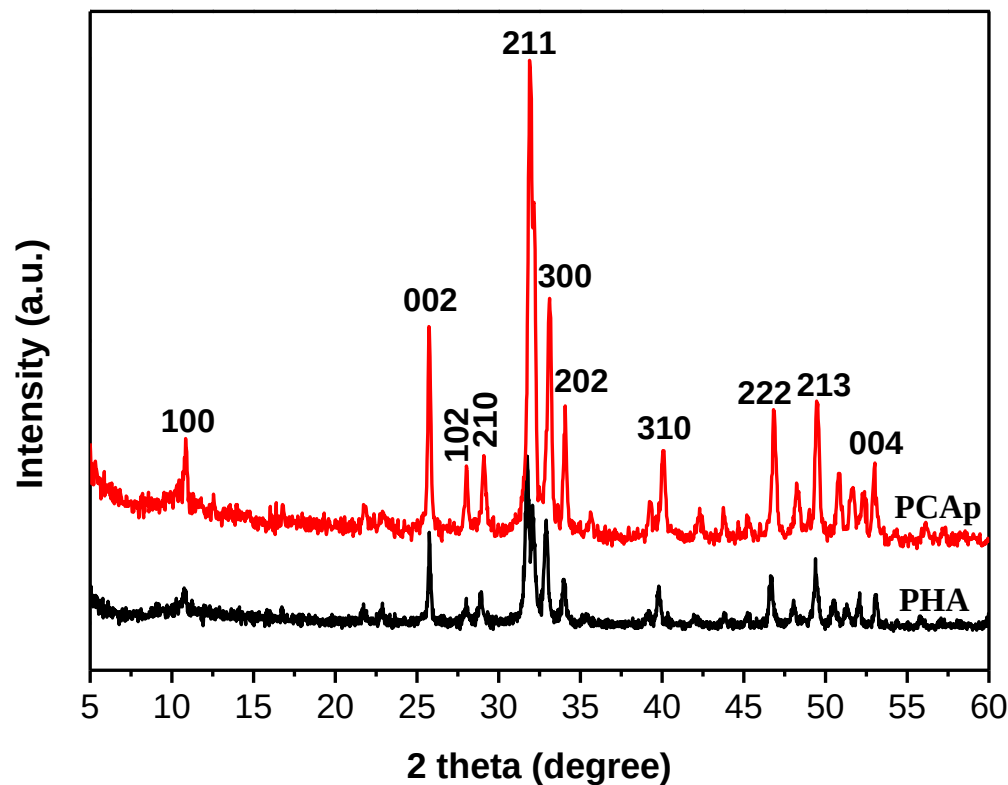
Lattice Parameter Analysis
Phase Fraction Analysis

- **Structure Analysis**

Crystal structure

- **Peak Shape Analysis**

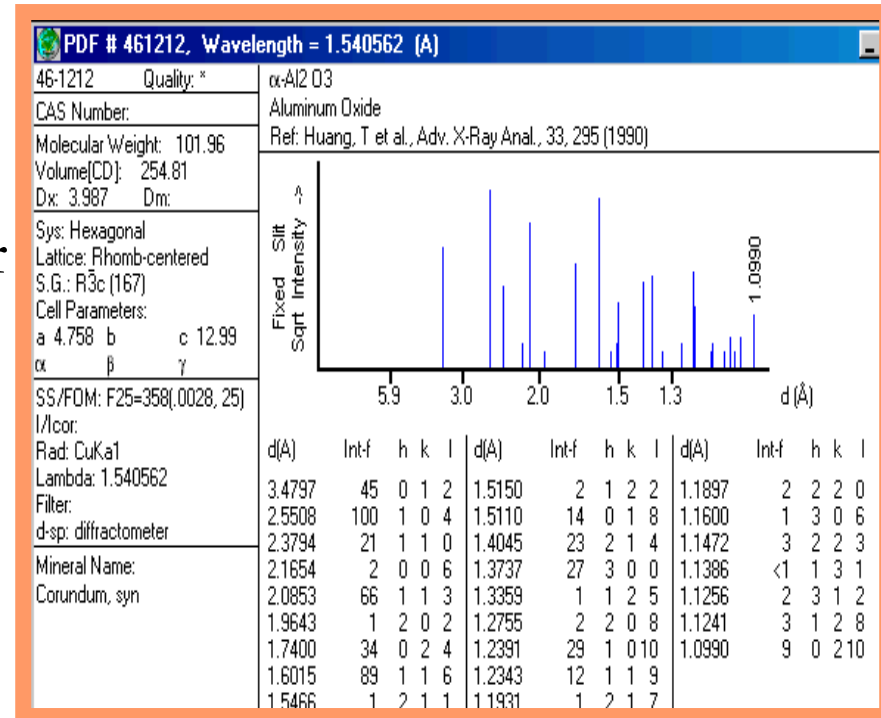
Crystallite Size Analysis
Microstrain Analysis



Phase Identification

- Comparison the diffraction pattern with reference patterns
- Database $\Rightarrow$ International Centre for Diffraction Data
- Diffraction file contains

diffraction pattern
molecular weight
structure
space group
lattice parameters



(<http://www.icdd.com>)

Size, Surface area, density, and thermal properties

Particle Size Analysis

Sieve analysis

Microscopy

Diffraction

Sedimentation



Specific Surface Area

BET technique



Density

Pycnometer



Thermal Properties

DTA / TGA

Sieve Analysis

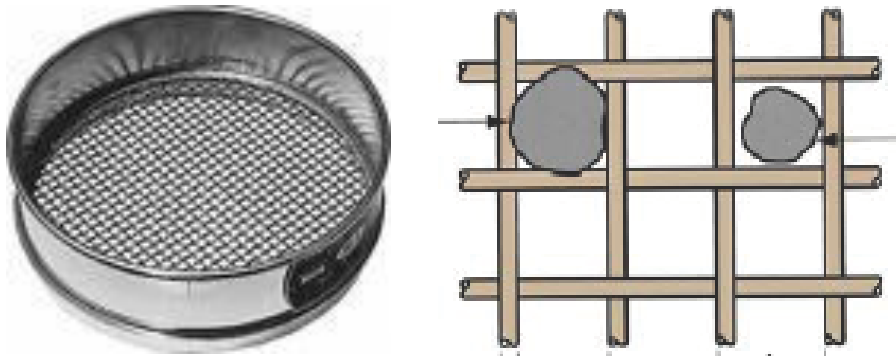
- Screens of different mesh sizes to capture different size particle
- Advantages:
 - variety of materials $\Rightarrow$ sands, clays, soil, etc.
 - simple & cost effective
- Drawbacks
 - inaccuracy for fine materials (<100 mesh) $\Rightarrow$ high surface attraction



(<http://www.mtsc.unt.edu/FACULTY/bgorman>)

Screen Size

- Mesh count $\Rightarrow$ # of openings per linear inch of screen
 - A mesh count of 200 $\Rightarrow$ 200 openings / linear inch
 - Higher mesh count $\Rightarrow$ smaller particle size



(<http://users.encs.concordia.ca>)

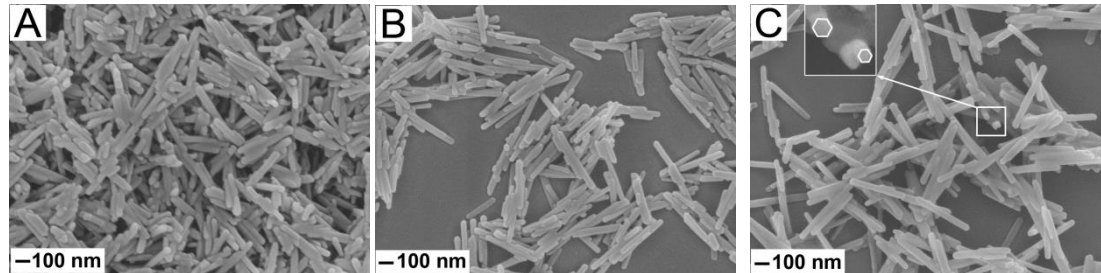
Sieve Number	Opening Size (mm)
4	4.750
6	3.350
8	2.360
12	1.680
16	1.180
20	0.850
30	0.600
40	0.425
50	0.300
60	0.250
80	0.180
100	0.150
140	0.106
200	0.075

Powder size, shape and morphology by SEM

- SEM displays shape and morphology of powders
- Maximum magnifications $\Rightarrow$ 50,000X to 500,000X depending on electron source
- Advantages of SEM

Rapid

Non-destructive



- Sputtered Coating: [F. Ren et al. Journal of Crystal Growth 2012;349: 75-82.](#)

Gold, gold/palladium alloy, platinum, tungsten, graphite

Image Analysis Software

- Common softwares for image analysis
 - ImageJ
 - Scion image
- How to use Scion image in particle size analysis ?
 - Import image
 - Measure scale bar
 - Set scale
 - Measure particle size
 - Import file to excel

Specific Surface Area by BET

$$\frac{1}{V_a[(P^o/P)-1]} = \frac{1}{V_m C} + \frac{(C-1)}{V_m C} \left(\frac{P}{P^o} \right)$$

where

V_a = volume of gas adsorbed at pressure P ;

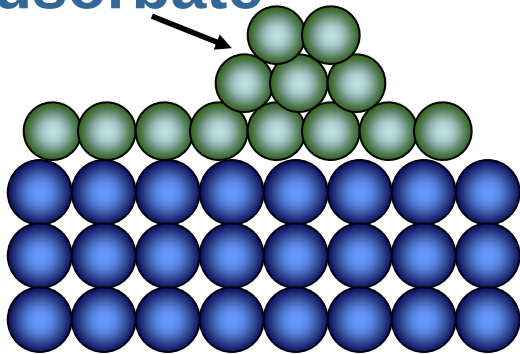
V_m = volume of gas required to form monolayer;

C = BET constant (related to energy of adsorption of 1st layer); and

$\frac{P}{P^o}$ = relative pressure of adsorbate.

S.Brunauer, P.Emmett, E.Teller Adsorption of Gases in Multimolecular Layers, J. Am. Chem. Soc., 1938, 60 (2), pp 309–319

Adsorbate



Adsorbent

- **Both monolayer and multilayer adsorption**
- **Assumptions:**
 - (a) **gas molecules physically adsorb on a solid in layers infinitely;**
 - (b) **there is no interaction between each adsorption layer;**
 - (c) **the Langmuir theory can be applied to each layer.**

Specific Surface Area Calculation

$$\frac{1}{V_a (P^o / P - 1)} = \frac{(C-1)}{V_m C} \left(\frac{P}{P^o} \right) + \frac{1}{V_m C}$$

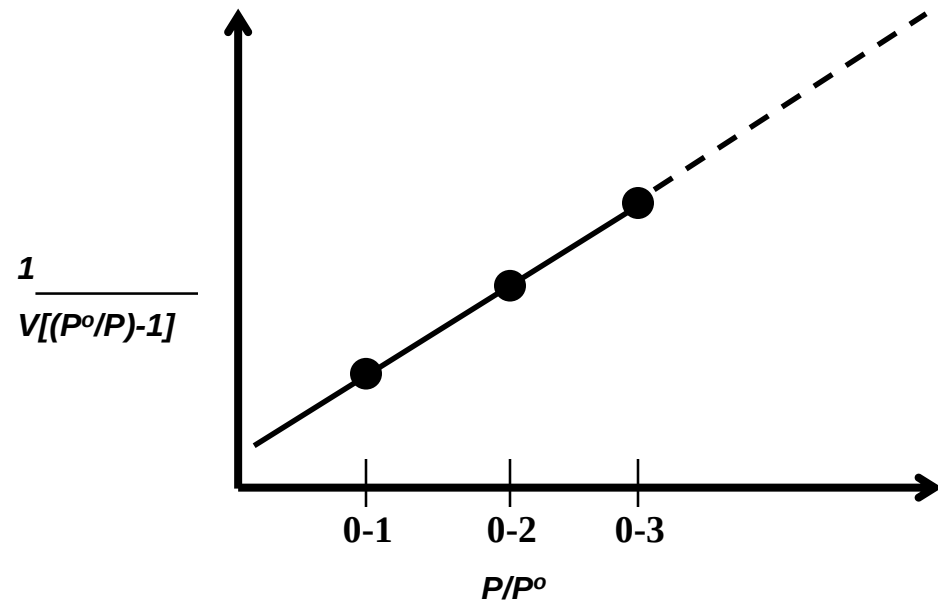
$$Y = mX + i$$

$$V_m = \frac{1}{m + i}$$

$$\text{Total surface area} = \frac{V_m N_{av} A_{cs}}{\text{Molar volume of adsorbate}}$$

$$\text{SSA (Specific surface area)} = \frac{\text{Total surface area}}{\text{Weight of sample}}$$

At least three data points in the relative pressure range 0.05 to 0.30



Processing of Advanced Ceramics

- ✦ Manufacturing sequence for advanced ceramics can be summarized in the following steps:
 1. Preparation of starting materials
 2. Shaping
 3. Sintering
 4. Finishing

- ✦ While the sequence is nearly the same as for the traditional ceramics, the details are often quite different

5.5 Processing of Advanced Ceramics

Shaping of Advanced Ceramics

- ✚ Many of the shaping processes are borrowed from powder metallurgy (PM) and traditional ceramics
 - 📄 PM press and sinter methods have been adapted to the advanced ceramic materials
- ✚ And some of the traditional ceramics forming techniques are used to shape the advanced ceramics
 - 📄 Slip casting
 - 📄 Extrusion
 - 📄 Dry pressing

Uniaxial Pressing

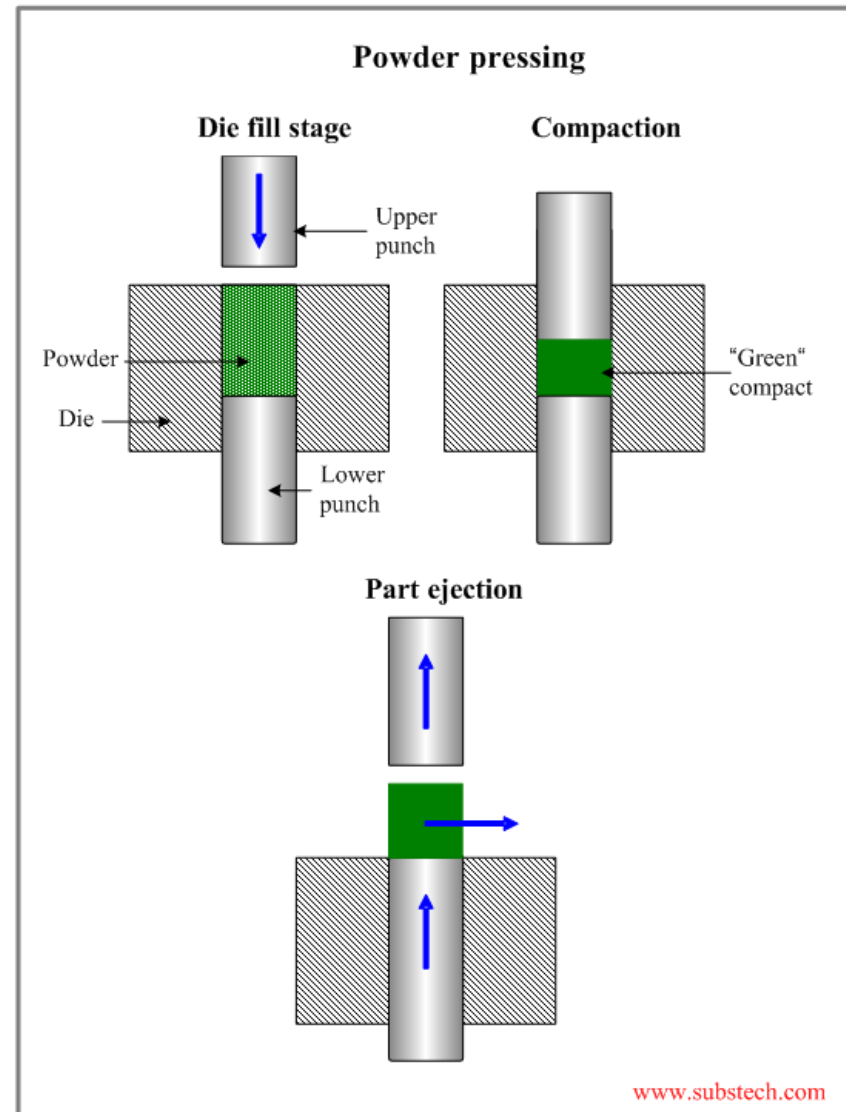
- Suitable for mass production of simple ceramics parts
- Referred to as Die Pressing
- Major components of simple dies
 - steel sleeve:
containing cavity
 - collar:
opened steel cylinder
 - pushing rods



(<http://www.mtixtl.com/>)

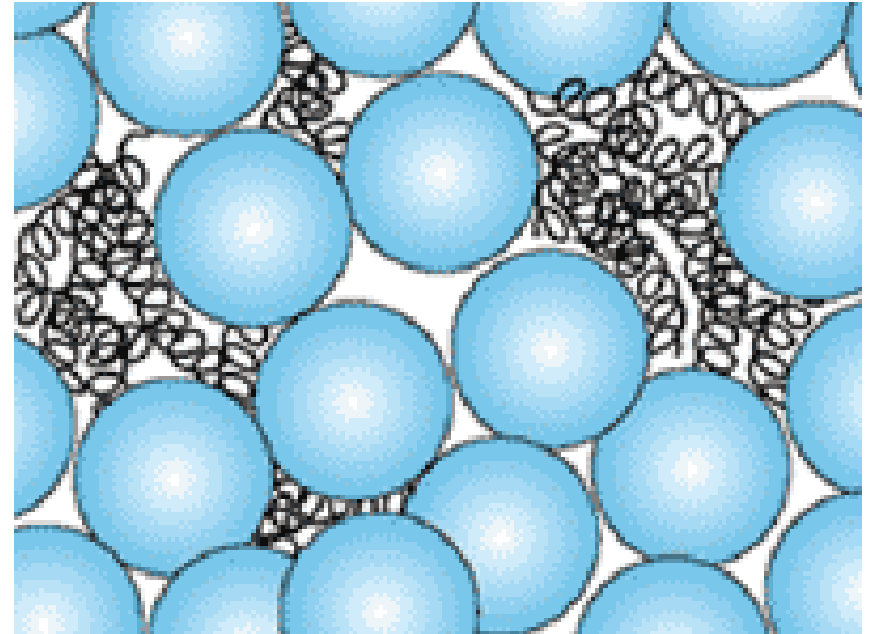
Uniaxial Pressing

- Powder pressed in closed dies $\Rightarrow$ green compact
- Processes involve:
 - filling of the die
 - compaction of powder
 - ejection of the ceramic part



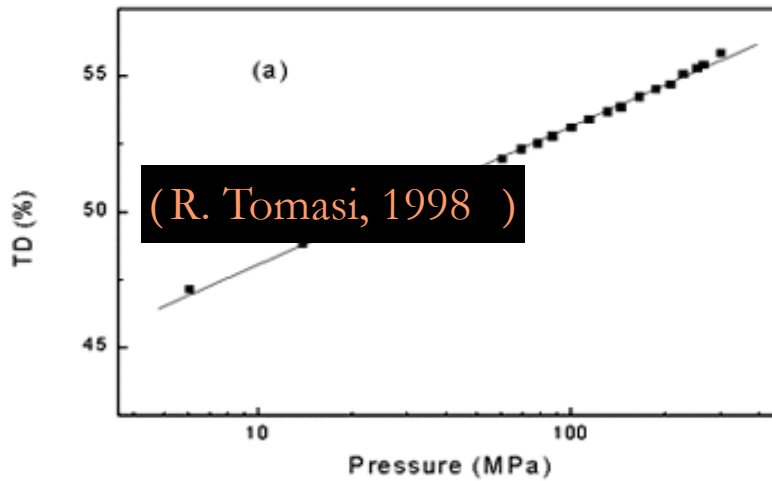
Die Filling

- Powder fed into die cavity
- Additives $\Rightarrow$ accommodate the pressing, improve the properties of Ceramic parts
- Types of additives:
 - binder
 - lubricant
 - plasticizer (增塑剂)



Powder Compaction

- Air removal $\Rightarrow$ press powder lightly
- Compaction $\Rightarrow$ move down upper punch, pressing powder with predetermined pressure
- Applied pressure: 69- 830 MPa



(www.pressmanpress.com/)

Hot Pressing

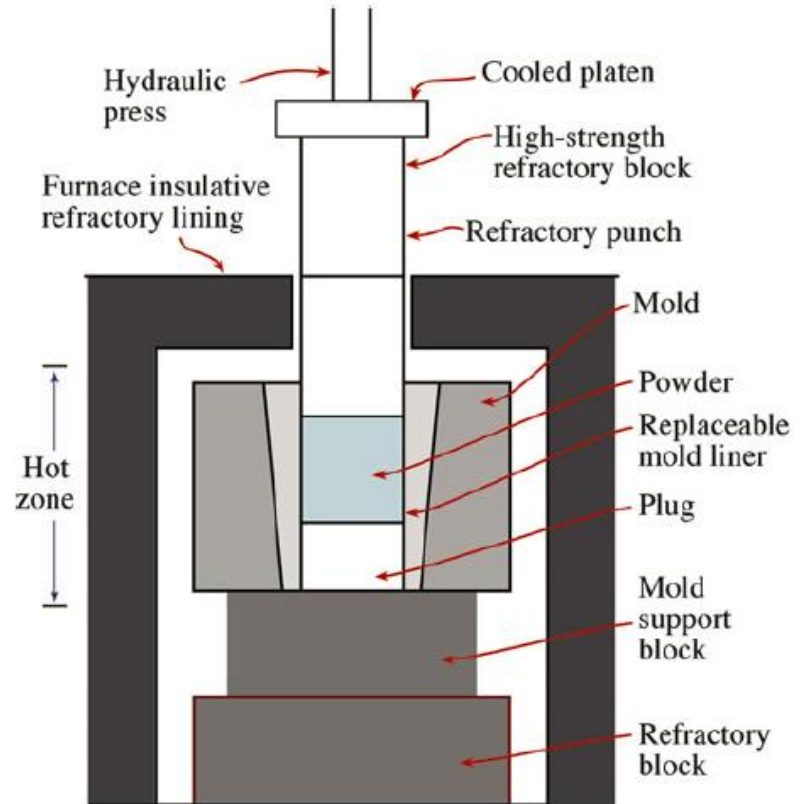
📌 Similar to dry pressing

📄 Sintering of the product is accomplished simultaneously with pressing

😊 Eliminates the need for a separate firing step

😊 Higher densities and finer grain size are obtained;

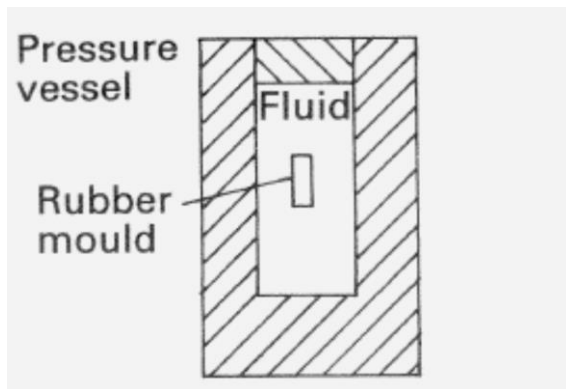
☹ Die life is reduced by the hot abrasive particles against the die surfaces



Isostatic Pressing

Uses hydrostatic pressure to compact the ceramic powders from all directions;

- ✦ Avoids the problem of non-uniform density in the final product that is often observed in conventional uniaxial pressing;
- ✦ Same process used in powder metallurgy



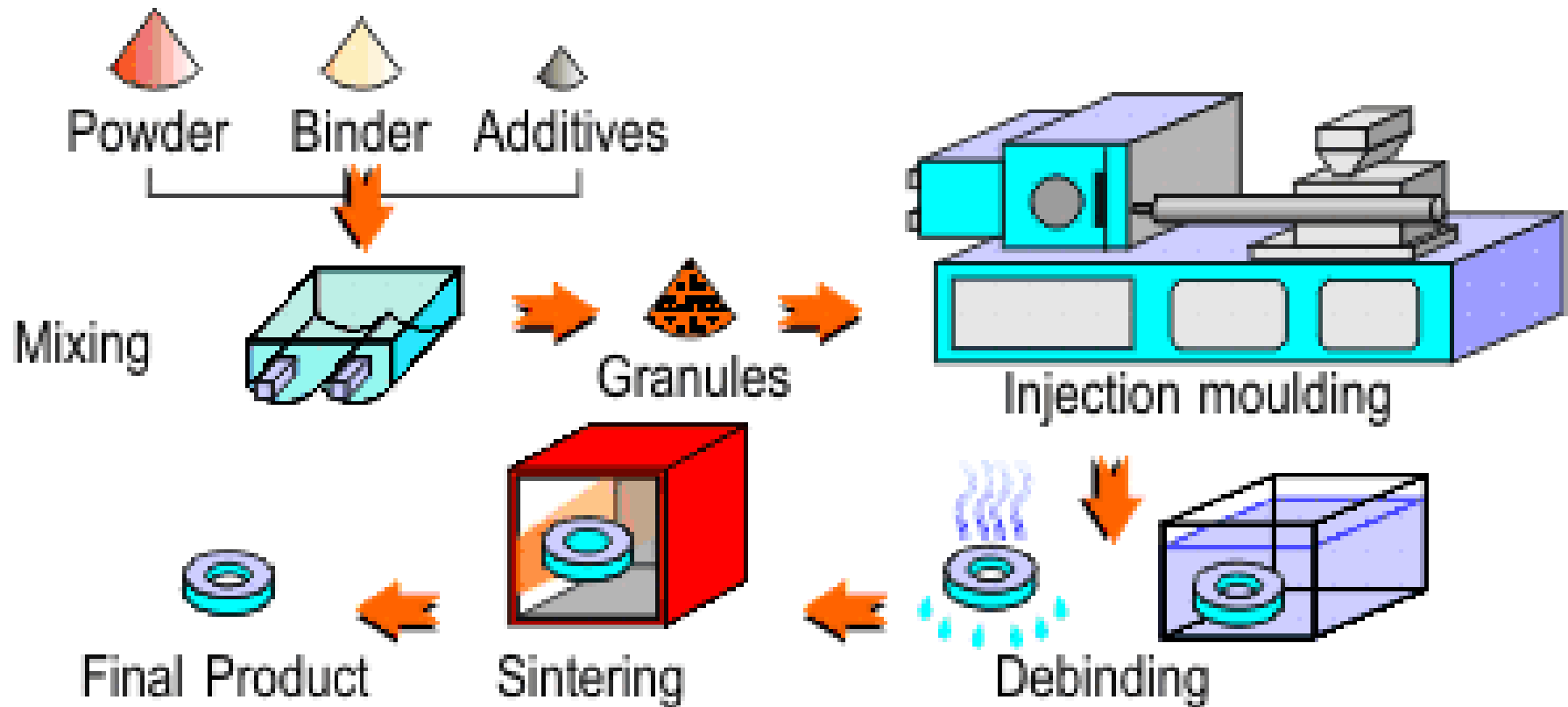
Uniform pressure gives uniform green density and limits lamination. Used for mass production of spark plugs and high-voltage insulators

Powder Injection Molding (PIM)

Ceramic particles are mixed with a thermoplastic, then heated and injected into a mold cavity;

- ✦ Polymer acts as a carrier and provides flow characteristics for molding;
- ✦ Upon cooling which hardens the polymer, the mold is opened and part is removed ;
- ✦ The plastic binder is removed and the remaining ceramic part is sintered .

Powder Injection Molding (PIM)



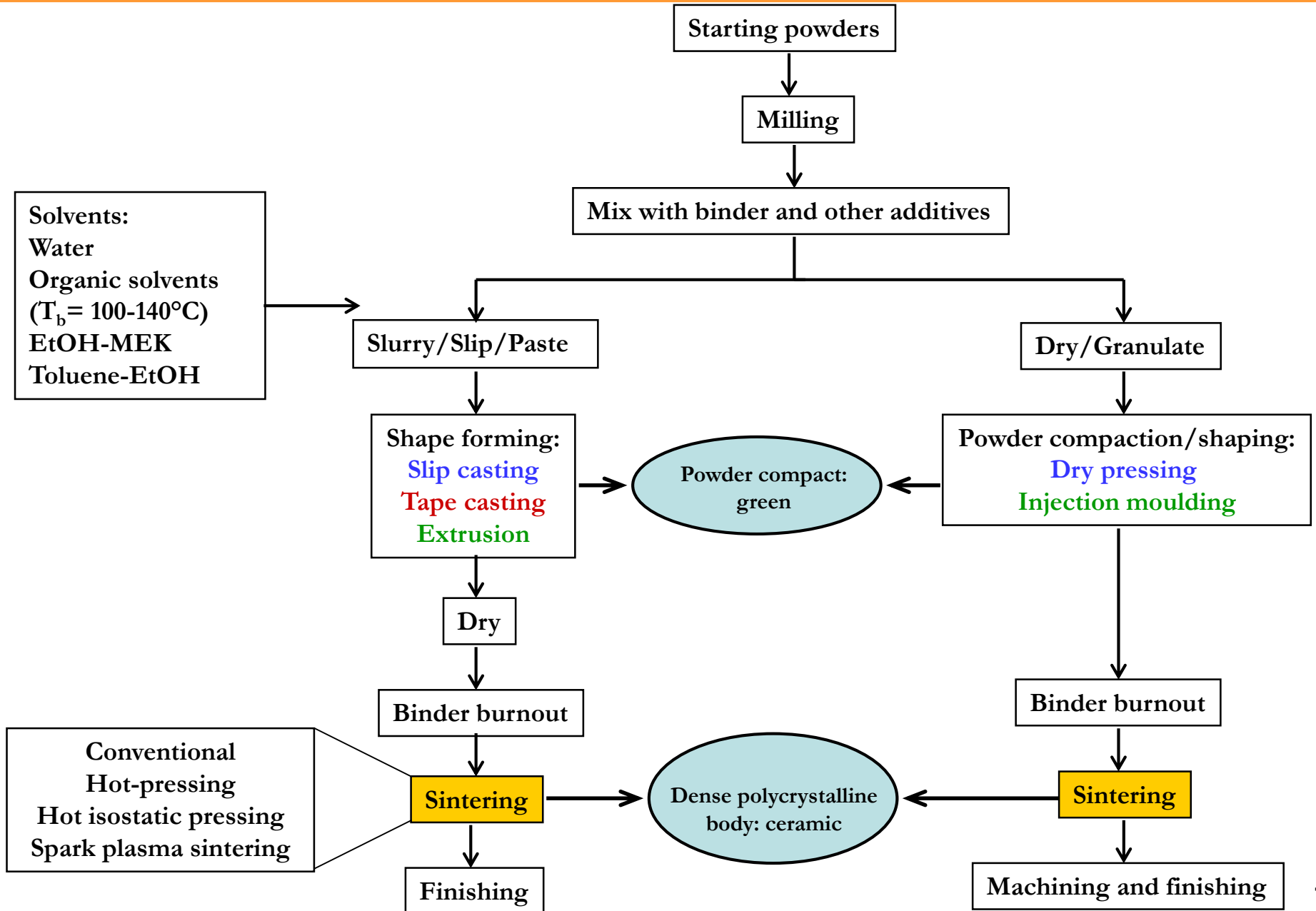
Sintering of Advanced Ceramics

- ✪ Since the plasticity needed to shape the new ceramics is not normally based on water, the drying step required for traditional green ceramics is omitted for most advanced ceramic products;
- ✪ Sintering step is still very much required;
- ✪ Functions of sintering are the same as before:
 1. Bond individual grains into a solid mass
 2. Increase density
 3. Reduce or eliminate porosity

Finishing Operations for Advanced Ceramics

- ✎ Parts made of advanced ceramics sometimes require finishing, with one or more of the following purposes:
 1. Increase dimensional accuracy
 2. Improve surface finish
 3. Make minor changes in part geometry
- ✎ Finishing usually involves abrasive processes
 - 📄 Diamond abrasives must be used to cut the hardened ceramic materials

Processing of Ceramics-Summary



Cemented Carbides

A family of composite materials consisting of carbide ceramic particles imbedded in a metallic binder

✎ Classified as metal matrix composites because the metallic binder is the matrix that holds the bulk material together

📄 However, the carbide particles constitute the largest proportion of the composite material

📖 Normally between 80% and 95% by volume

Binders for Cemented Carbides

- ✚ Carbide powders must be sintered with a metal binder to provide a strong and pore-free part
 - 📄 Cobalt works best with WC
 - 📄 Percentage of binder metal is 4% up to 20%
- ✚ Powders of carbide and binder are thoroughly mixed wet in a ball mill to form a homogeneous sludge
- ✚ The sludge is then dried in a vacuum or controlled atmosphere to prevent oxidation before compaction

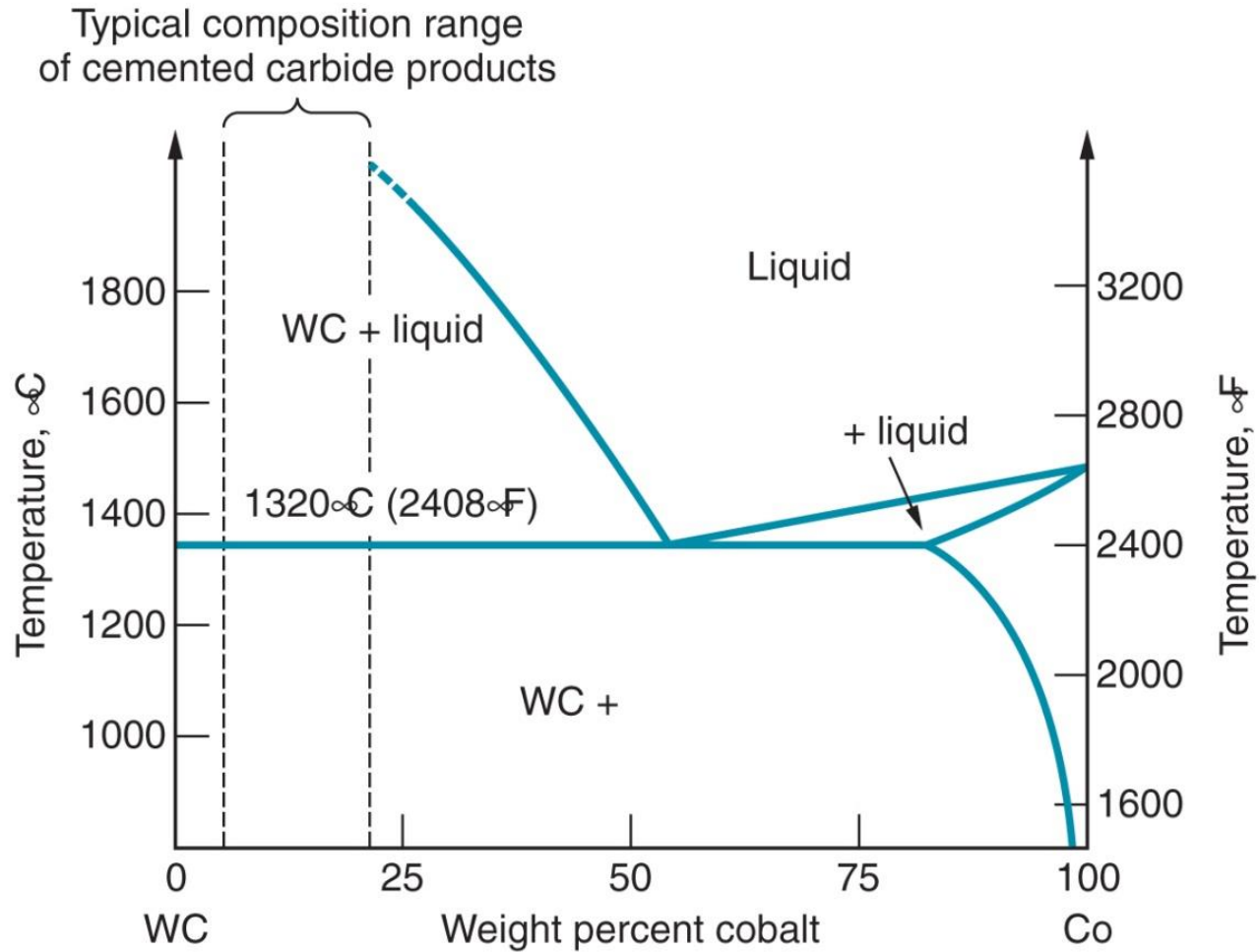
Compaction

- ✎ Most common process is cold pressing, used for high production of cemented carbide parts such as cutting tool inserts
- 📄 Dies must be oversized to account for shrinkage during sintering
- 📄 For high production, dies are made with WC-Co liners to reduce wear
- 📄 For smaller quantities, large flat sections may be pressed and then cut into smaller pieces

Sintering of WC-Co

- ✦ Possible to sinter WC without a metal binder, but the resulting material is less than 100% of true density
 - 📄 Using a binder yields a structure virtually free of porosity
- ✦ Sintering of WC-Co = liquid phase sintering
 - 📄 Usual sintering temperatures for WC-Co are 1370-1425°C , which is below cobalt's melting point of 1495°C ;Thus, the pure binder metal does not melt at the sintering temperature

WC-Co Phase Diagram



Sintering of WC-Co

- ✎ WC is gradually dissolved in Co during sintering, and its melting point is reduced so melting does occur
 - 📄 As liquid phase forms, it flows and wets the WC particles, further dissolving the solid
 - 📄 Molten metal also serves to remove gases from the internal regions of the compact
- ✎ These mechanisms cause rearrangement of the remaining WC particles into a closer packing
 - 📄 Results in significant densification and shrinkage of the WC-Co mass

Secondary Operations

- ✦ Subsequent processing is usually required after sintering to achieve adequate dimensional control of the cemented carbide parts
- ✦ Grinding with a diamond or other very hard abrasive wheel is the most common secondary operation performed for this purpose
- ✦ Other secondary operations for shaping include
 - 📄 Electric discharge machining
 - 📄 Ultrasonic machining